8 DE AGOSTO DE 2016

1) $H_c(s) = \dfrac{s+3}{(s+1)^2+2}$

$(s+1)^2 + 2 = 0$

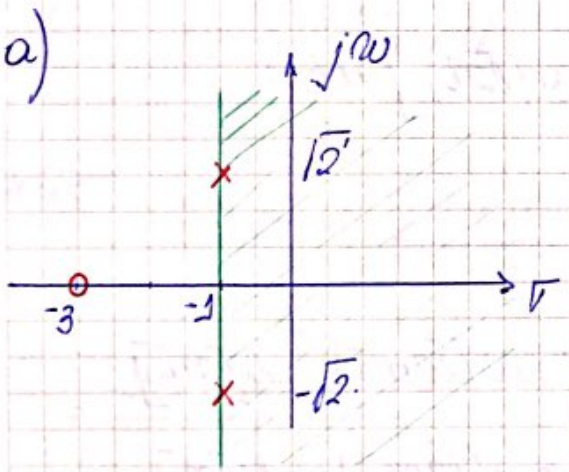
$s^2 + 2s + 1 + 2 = 0$

$s^2 + 2s + 3 = 0 \Rightarrow s_{1,2} = \dfrac{-2 \pm i\sqrt{8}}{2}$

$s_{1,2} = \dfrac{-2 \pm i2\sqrt{2}}{2} = -1 \pm i\sqrt{2}$

$\Rightarrow H_c(s) = \dfrac{s+3}{(s-[-1+i\sqrt{2}])(s-[-1-i\sqrt{2}])}$

a)



CAUSAL $\Rightarrow$ LIMITADA A IZQUIERDA

ESTABLE $\Rightarrow$ INCLUYE AL EJE jw

ROC: $\mathrm{Re}[s] > -1$

$\sigma > -1$

b) $H_c(s) = \dfrac{s+1+2}{(s+1)^2+2} = \dfrac{s+1}{(s+1)^2+2} + \dfrac{2}{(s+1)^2+2}$

$H_c(s) = \dfrac{s+1}{(s+1)^2+(\sqrt{2})^2} + \dfrac{\sqrt{2}\sqrt{2}}{(s+1)^2+(\sqrt{2})^2}$

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$$e^{-\alpha t}\cos(\omega_0 t)\,u(t) \longrightarrow \frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$$

$$\frac{s+1}{(s+1)^2+(\sqrt{2})^2} \rightarrow e^{-t}\cos(\sqrt{2}t)\,u(t)$$

$$e^{-\alpha t}\,\text{SEN}(\omega_0 t)\,u(t) \longrightarrow \frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$$

$$\frac{\sqrt{2}\sqrt{2}}{(s+1)^2+(\sqrt{2})^2} \longrightarrow e^{-t}\sqrt{2}\,\text{SEN}(\sqrt{2}t)\,u(t)$$

ENTONCES

$$h_c(t) = e^{-t}\left[\cos(\sqrt{2}t) + \sqrt{2}\,\text{SEN}(\sqrt{2}t)\right]u(t)$$

SEA:

$$h_d[n] = h_c(nT) \;;\; T>0.$$

$$h_d[n] = e^{-nT}\left[\cos(\sqrt{2}nT) + \sqrt{2}\,\text{SEN}(\sqrt{2}nT)\right]u[nT]$$

$$h_d[n] = e^{-nT}\left[\cos(\sqrt{2}nT) + \sqrt{2}\,\text{SEN}(\sqrt{2}nT)\right]u[n]$$

$$r^n\cos(\omega_0 n)\,u[n] \longrightarrow \frac{1-[r\cos(\omega_0)]z^{-1}}{1-[2r\cos(\omega_0)]z^{-1}+r^2z^{-2}}$$

$$r^n\,\text{SEN}(\omega_0 n)\,u[n] \rightarrow \frac{r\,\text{SEN}(\omega_0)\,z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$$

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$$Hd(z) = \frac{1 - e^{-T}\cos(\sqrt{2}T)\,z^{-1}}{1 - 2e^{-T}\cos(\sqrt{2}T)\,z^{-1} + e^{-2T}z^{-2}} + \sqrt{2}\,\frac{e^{-T}\,\text{SEN}(\sqrt{2}T)\,z^{-1}}{1 - 2e^{-T}\cos(\sqrt{2}T)\,z^{-1} + e^{-2T}z^{-2}}$$

$$Hd(z) = \frac{1 - e^{-T}\cos(\sqrt{2}T)\,z^{-1} + \sqrt{2}e^{-T}\,\text{SEN}(\sqrt{2}T)\,z^{-1}}{1 - 2e^{-T}\cos(\sqrt{2}T)\,z^{-1} + e^{-2T}z^{-2}}$$

$$\text{Roc: } |z| > e^{-T}$$

$$Hd(z) = \frac{z\left\{z - \left[e^{-T}\cos(\sqrt{2}T) - \sqrt{2}e^{-T}\,\text{SEN}(\sqrt{2}T)\right]\right\}}{z^2 - 2e^{-T}\cos(\sqrt{2}T)\,z + e^{-2T}}$$

Ceros: $C_1 = 0$

$$C_2 = e^{-T}\left[\cos(\sqrt{2}T) - \sqrt{2}\,\text{SEN}(\sqrt{2}T)\right]$$

Polos: $$\frac{2e^{-T}\cos(\sqrt{2}T) \pm \sqrt{4e^{-2T}\cos^2(\sqrt{2}T) - 4e^{-2T}}}{2} =$$

$$= \frac{2e^{-T}\cos(\sqrt{2}T) \pm \sqrt{4e^{-2T}(1 - \text{SEN}^2(\sqrt{2}T)) - 4e^{-2T}}}{2}$$

$$= \frac{2e^{-T}\cos(\sqrt{2}T) \pm \sqrt{-4e^{-2T}\,\text{SEN}^2(\sqrt{2}T)}}{2}$$

$$= e^{-T}\cos(\sqrt{2}T) \pm j\,e^{-T}\,\text{SEN}(\sqrt{2}T)$$

$$= e^{-T}e^{\pm j\sqrt{2}T} = P_{1,2}$$

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$\Rightarrow |p_{1,2}| = e^{-\bar{T}} < 1 \Rightarrow$ EL SISTEMA ES ESTABLE Y CAUSAL!

~ PARA CUALQUIER VALOR QUE SE ELIJA DE $\bar{T}$, EL SISTEMA RESULTA CAUSAL Y ESTABLE

3) $$X[\bar{K}] = \sum_{m=0}^{N-1} x[m]\, W_N^{mK} = \sum_{m=0}^{N/2-1} x[m]\, W_N^{Km} + \sum_{m=N/2}^{N-1} x[m]\, W_N^{Km}$$

CAMBIO DE VARIABLE

$m = m - N/2$ ↘ $m = N-1 \Rightarrow m = N/2-1$

↘ $m = N/2 \Rightarrow m = 0$

↓

$m = m + N/2$

$$X[\bar{K}] = \sum_{m=0}^{N/2-1} x[m]\, W_N^{Km} + \sum_{m=0}^{N/2-1} x[m+N/2]\, W_N^{(m+N/2)K}$$

$$X[\bar{K}] = \sum_{m=0}^{N/2-1} x[m]\, W_N^{Km} + \sum_{m=0}^{N/2-1} x[m+N/2]\, W_N^{Km}\, e^{-j\bar{\pi}K}$$

$$X[\bar{K}] = \sum_{m=0}^{N/2-1} x[m]\, W_N^{Km} + \sum_{m=0}^{N/2-1} x[m+N/2]\, (-1)^K\, W_N^{Km}$$

$$X[\bar{K}] = \sum_{m=0}^{N/2-1} \left[x[m] + (-1)^K x[m+N/2] \right] W_N^{Km} \quad ; K = 0, \dots, N-1$$

PARES:

$$X[2\bar{K}] = \sum_{m=0}^{N/2-1} \left[x[m] + x[m+N/2] \right] W_N^{2Km}$$

$$X[2\bar{K}] = \sum_{m=0}^{N/2-1} \underbrace{\left[x[m] + x[m+N/2] \right]}_{\tilde{x}[m]} W_{N/2}^{Km} \quad ; K = 0, \dots, \frac{N}{2}-1$$

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IMPARES

$$X[2K+1] = \sum_{m=0}^{N/2-1} [x[m] - x[m+N/2]] W_N^{(2K+1)m}$$

$$X[2K+1] = \sum_{m=0}^{N/2-1} [x[m] - x[m+N/2]] W_{N/2}^{mK} W_N^{m}$$

$$X[2K+1] = \sum_{m=0}^{N/2-1} \underbrace{[x[m] - x[m+N/2]] W_N^{m}}_{\hat{x}[m]} W_{N/2}^{mK} \; ; \; K = 0, \cdots, \frac{N}{2}-1$$

ENTONCES:

$$\tilde{x}[m] = x[m] + x[m+N/2]$$

$$\hat{x}[m] = \{x[m] - x[m+N/2]\} W_N^{m} \; ; \; W_N^{m} = e^{-j\frac{2\pi}{N}m}$$

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